Pham Hung Quoc Viet

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OBJECTIVE

Analytical 3rd-year student at University of Information Technology – VNU-HCM (UIT) with a solid foundation in Probability & Statistics, Time-Series Analysis, and Machine Learning. Experienced in building high-throughput stream processing pipelines, engineering rolling-window statistical features, and benchmarking predictive models using Python and PySpark. Seeking a Quantitative Trading / Fintech Intern position to apply mathematical, statistical, and algorithmic modeling to develop, backtest, and optimize data-driven trading strategies.

EDUCATION

University of Information Technology – VNU-HCM (UIT) 2023 – 2027 (expected)
Bachelor of Information Systems GPA: 8.06/10
Relevant Coursework: Probability & Statistics, Applied Statistics, Machine Learning, Data Mining, Database Management Systems

SKILLS

Programming	Python, SQL, C++, Java, PHP
Quantitative & Stats	Probability & Statistics, Time-Series Analysis, Statistical Modeling, EDA, Data Wrangling
AI & Machine Learning	Scikit-learn, XGBoost, LightGBM, Random Forest, Anomaly Detection, Feature Engineering, Explainable AI (XAI), Class Imbalance Handling, Optuna Tuning, TensorFlow
Data & Big Data	Pandas, NumPy, Matplotlib, Seaborn, Apache Spark (PySpark), Kafka, SQL Server, PostgreSQL
Tools & Platforms	Git, GitHub, Docker, Streamlit, Jupyter Notebook, VS Code, Excel, SSIS
Others & Languages	IELTS 6.5 (2022), Agile/Scrum Methodology

PROJECTS

NYC Taxi — Real-time Fleet Intelligence & Anomaly Detection 2026
PySpark, Kafka, Docker, PostgreSQL, Streamlit, Pandas, Scikit-learn | GitHub

- **Data Ingestion:** Developed low-latency Kafka ingestion pipeline; simulated high-frequency event producer streaming message streams at 200 msg/s.
- **Time-Series Feature Engineering:** Utilized PySpark Structured Streaming (5,000 offsets/trigger) to compute 14 rolling-window statistical features and moving averages using Pandas.
- **Anomaly & Pattern Detection:** Designed an Isolation Forest and Minimum Covariance Determinant (MinCovDet) ensemble with strict voting to identify statistical anomalies and regime shifts.
- **Deployment & Retraining:** Managed continuous ingestion to PostgreSQL; implemented automated model evaluation and parameter retraining triggered every 25 streaming batches.

Bank Customer Churn Prediction & Explainability 2026
Python, Scikit-learn, TensorFlow, XGBoost, LightGBM, SMOTE, SHAP, Optuna, Pandas, Seaborn | GitHub

- **Statistical Preprocessing:** Conducted extensive exploratory analysis on a bank dataset; applied Winsorization (outlier mitigation), encoding, and normalization to improve the signal-to-noise ratio.
- **Class Balancing:** Implemented SMOTE to resolve class imbalance (20.4% positive rate), statistically augmenting data from 6,400 to 10,192 samples to enhance model sensitivity.
- **Predictive Model Benchmarking:** Trained, optimized, and evaluated 7+ machine learning algorithms (XGBoost, Random Forest, LightGBM, etc.) using Optuna and GridSearchCV for hyperparameter tuning.
- **Model Explainability & Risk Attribution:** Utilized SHAP and LIME to dissect feature contributions; employed Jaccard similarity analysis to quantify stability of key explanatory variables.

E-commerce Consumer Behaviour Analysis & BI Solution 2025
SSIS, SSAS, SQL Server, Power BI, MDX, Python, Scikit-learn | GitHub

- **Data Pipeline:** Engineered automated SSIS pipelines to clean, pre-process, and consolidate high-volume data streams into SQL Server.
- **Dimensional & Statistical Modeling:** Modeled a structured Star Schema and built OLAP cubes with MDX to perform hierarchical slice-and-dice data analysis.
- **Quantitative Segmentation:** Implemented K-Means clustering for behavioral segmentation and Decision Tree classification to predict customer spending levels using Python (Scikit-learn).
- **BI Visualization:** Designed interactive dashboards combining model predictions and business metrics to track key performance indicators.